Assignment 2 (02/21/2026)

Finite Difference Approximation (1D case)

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The purpose of this problem is to determine approximation formulas for high order derivatives and their order of the error terms. Please provide detailed solutions to these problems.

Problem 1

(1.0 pts.) Use the fourth order Taylor series to approximate $\cos(0.2)$ around zero. Determine

- the exact error after every iteration: after adding zero, first, second, third and fourth order terms
- the truncation error bound: the leading term after truncation.

Make a plot to compare both errors, and discuss their behaviors. You can use the example code given in pages 320-321, Ch. 18 of the book as a template.

Problem 2

(a) (1.0 pts.) Use Taylor series to show the following approximations and their accuracy

(i) $f''(x_j) = \frac{-f(x_{j+3})+4f(x_{j+2})-5f(x_{j+1})+2f(x_j)}{h^2} + O\left(h^2\right)$,

(ii) $f'''(x_j) = \frac{f(x_{j+3})-3f(x_{j+2})+3f(x_{j+1})-f(x_j)}{h^3} + O(h)$

(b) (1.0 pts.) Use Taylor series to verify that the central difference formula

$$f'(x_j) \approx \frac{f(x_{j+1}) - f(x_{j-1})}{2h}$$

has error of order $O\left(h^2\right)$.

(c) (1.0 pts.) Compute the numerical derivative of $f(x) = \cos(x)$ and plot the maximum error between the approximated and the exact derivatives for $x \in [0, 2\pi]$ vs. h (check page 344 Ch. 20 in the book).

(d) (1.0 pts.) Suppose we have a function

$$f(x) = x^2$$

and we want to approximate the derivative at x_0 in the interval $[0, 4]$ with a step size $h < 0.5$. Using the central difference scheme, show that this approximation is exact (i.e., the error is zero).